SNOWPRO® ADVANCED: DATA ENGINEER (DEA-C02) EXAM STUDY GUIDE

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SNOWPRO ADVANCED: DATA ENGINEER STUDY GUIDE OVERVIEW

This study guide highlights concepts that may be covered on Snowflake's SnowPro Advanced: Data Engineer Certification exam.

Use of this study guide does not guarantee certification success.

Holding the SnowPro Core certification in good standing is a prerequisite for taking the Advanced: Data Engineer certification.

For an overview and more information on the SnowPro Core Certification exam or SnowPro Advanced Certification series, please navigate here.

RECOMMENDATIONS FOR USING THE GUIDE

This guide will show the Snowflake topics and subtopics covered on the exam. Following the topics will be additional resources consisting of videos, documentation, blogs, and/or exercises to help you understand data engineering with Snowflake.

Estimated length of time required to complete the study guide: 10 – 13 hours.

Some links may have more value than others, depending on your experience. The same amount of time should not be spent on each link. Some links may appear in more than one content domain.

SNOWPRO ADVANCED: DATA ENGINEER CERTIFICATION OVERVIEW

The **SnowPro Advanced: Data Engineer** certification tests advanced knowledge and skills used to apply comprehensive data engineering principles using Snowflake. The exam will assess skills through scenario-based questions and real-world examples.

This certification will test the ability to:

- Source data from Data Lakes, APIs, and on-premises
- Transform, replicate, and share data across cloud platforms
- Design end-to-end near real-time streams
- Design scalable compute solutions for Data Engineer workloads
- Evaluate performance metrics

Target Audience:

2 + years of data engineering experience, including practical experience using Snowflake for Data Engineer tasks. In addition, successful candidates may have:

- A working knowledge of Restful APIs, SQL, semi-structured and unstructured data sets, and cloud native concepts.
- Programming experience is a plus.

This exam is designed for:

- Data Engineers
- Software Engineers
- Technical Data Leads
- Lead System Engineers

SNOWPRO ADVANCED: DATA ENGINEER PREREQUISITE

Eligible individuals must hold an active SnowPro Core credential. If you feel you need more guidance on the fundamentals, please see the SnowPro Core Exam Study Guide.

STEPS TO SUCCESS

1. Review the Data Engineer Exam Guide
2. Attend Snowflake's Instructor-Led Data Engineering I Training
3. Attend Snowflake's Instructor-Led Data Engineer II Training
4. Review and study applicable white papers and documentation
5. Review and study the Snowflake Well Architected Framework documentation
6. Get hands-on experience with relevant business requirements using Snowflake
7. Attend Snowflake Webinars
8. Attend Snowflake Virtual Hands-on Labs for practical experience
9. Take advantage of Snowflake Developer Guides
10. Take advantage of Snowflake Northstar hands-on Data Engineering instruction
11. Take an official Snowflake Practice exam here
12. Schedule your exam
13. Take your exam!

Check out this free Snowflake Data Engineering tool:

Cloud Data Engineering for Dummies

SNOWPRO ADVANCED: DATA ENGINEER SUBJECT AREA BREAKDOWN

The following table contains the domains and weightings covered on the exam. It is not a comprehensive listing of all the content that will be presented on the exam.

Domain	Domain Weightings
1.0 Data Movement	28%
2.0 Performance Optimization	19%
3.0 Storage and Data Protection	14%
4.0 Data Governance	14%
5.0 Data Transformation	25%

Domain 1.0: Data Movement

1.1 Given a data set, load data into Snowflake.

- Outline considerations for data loading
- Define data loading features and potential impacts

1.2 Ingest data of various formats through the mechanics of Snowflake.

- Required file formats
- Schema detection using INFER_SCHEMA for table design and data analysis
- Ingestion of structured, semi-structured, and unstructured data
- Implementation of stages and file formats
  - Manage storage integrations configurations
  - Manage encryption (pre-scoped URLs, server-side, or client-side)
  - Manage compression and parsing strategies
- Extract metadata from staged files

1.3 Troubleshoot data ingestion.

- Identify causes of ingestion errors
- Determine resolutions for ingestion errors

1.4 Design, build, and troubleshoot continuous data pipelines.

- Stages
- Tasks
- Streams
- Dynamic tables
- Materialized views
- Snowpipe (for example, Auto Ingest compared to the REST API)
- Snowpipe Streaming
  - Snowpipe Streaming compared to the Kafka connector
- Create User-Defined Functions (UDFs)
- Design and use the Snowflake SQL API
- Openflow
- Use Notebooks to run pipelines of stored procedures for data ingestion tasks
- Use Snowflake scripting to develop and automate pipelines

1.5 Install, configure, and use connectors for Snowflake integration.

- Kafka connectors
- Spark connectors
- Python connectors
- Native connectors

1.6 Design and build data sharing and data consumption solutions.

- Evaluate the use of a data share or a clone
- Implement a data share
  - Manage auto-fulfillment
- Create and manage views
- Implement row-level filtering
- Share data using the Snowflake Marketplace
- Share data using a listing
- Use Streamlit to build data applications and interfaces for data consumption
  - Create interactive dashboards for data exploration and sharing
  - Build self-service data access applications

1.7 Manage different types of tables and data operations.

- Manage external tables
- Manage Iceberg tables
- Manage hybrid tables
- Perform general table management
- Use Horizon Catalog to federate data from external catalogs
- Manage schema evolution
- Unload data

Domain 1.0 Study Resources

Lab Guides

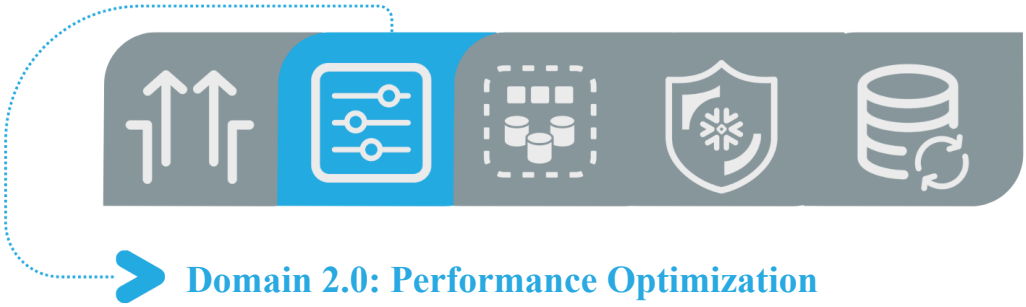
Accelerating Data Engineering with Snowflake & dbt
Auto-Ingest Twitter Data into Snowflake
Automating Data Pipelines to Drive

Additional Assets

Moving from On-Premises ETL to Cloud-Driven ELT (white paper)
Intro to Data Engineering with Snowpark for Python (virtual hands-on lab)
Building Continuous Data Pipelines with Snowflake (virtual hands-on labs

Snowflake Documentation Links

Bulk loading - Local File Systems
COPY_HISTORY
COPY INTO
CREATE EXTERNAL TABLE
CREATE FILE FORMAT
CREATE STREAM
CREATE TASK
Data Loading Tutorials
Data loading / unloading DDL
Data Sharing
Databases, Tables & Views
DESCRIBE STAGE
Dynamic Tables
External Tables
File Functions
HASH
Internal stage parameters
Kaffka and Spark Connectors
Loading Data into Snowflake
Preparing to Load Data
Python Connector
Schema Detection with INFER_SCHEMA
SYSTEM$PIPE_STATUS
Snowflake SQL Data Types
Snowpipe
Snowpipe REST API
Snowpipe Streaming
Streams and Tasks
VALIDATE - Data loading
VALIDATE_PIPE_LOAD



Domain 2.0: Performance Optimization

2.1 Troubleshoot underperforming queries.

- Identify underperforming queries
- Outline telemetry around the operation
- Identify the root cause
- Increase efficiency

2.2 Given a scenario, configure a solution for optimal performance.

- Scale out compared to scale up
- Virtual warehouse properties (for example, size, multi-cluster)
  - Snowpark-optimized virtual warehouses
- Query complexity
- Micro-partitions and the impact of clustering
- Materialized views
- Search optimization service
- Query acceleration service
- Snowpark-optimized warehouses
- Caching features
- Use the ACCOUNT_USAGE schema
- Use warehouse metrics (such as warehouse queues) and configurations:
  - Resource monitors
  - Warehouse constraints on credit consumption
- Balance optimization with credit consumption considerations
- Optimize storage configurations and costs

2.3 Monitor continuous data pipelines.

- Snowflake objects
  - Tasks
    - Snowsight task history dashboards
  - Streams
  - Snowpipe Streaming
  - Alerts
  - Dynamic Tables
- Notifications
- Data quality and data metric function monitoring

Domain 2.0 Study Resources

Lab Guides

Resource Optimization: Performance
Resource Optimization: Usage Monitoring

Additional Assets

Performance Impact from Local and Remote Disk Spilling (blog)
Snowflake: Visualizing Warehouse Performance (blog)
Virtual Warehouse Best Practices (blog)

Snowflake Documentation Links

Account Usage
Configuring tasks to send notifications
COPY_HISTORY
Data quality and data metric functions
Databases, Tables & Views
LOAD_HISTORY View
Monitor query activity with QUERY HISTORY
Monitoring data loading
PIPE_USAGE_HISTORY view
Queries
QUERY_HISTORY,QUERY_HISTORY_BY_* SHOW STREAMS
System clustering information
System functions
SYSTEM$CLUSTERING_INFORMATION
TASK_HISTORY
Using persisted query results
Using the query acceleration Service
Virtual Warehouses

Domain 3.0: Storage & Data Protection

3.1 Implement and manage data recovery features in Snowflake.

- Time Travel
  - Impact of streams
- Fail-safe
- Cross-region and cross-cloud replication

3.2 Use system functions to analyze micro-partitions.

- Clustering depth
- Cluster keys
- Automatic Clustering features and optimizations

3.3 Use Time Travel and cloning to create new development environments.

- Clone objects
  - Permission inheritance
- Validate changes before promoting
- Rollback changes

Domain 3.0 Study Resources

Lab Guides

Getting Started with Time Travel

Snowflake Documentation Links

Cloning considerations
Clustering keys and clustered tables
CREATE STREAM
Database replication and failover/failback
Account replication
Micro-partitions and data clustering
Parameters
Snowflake Time Travel & Fail-safe
Stage, pipe, and load history replication
SYSTEM$CLUSTERING_DEPTH
SYSTEM$CLUSTERING_INFORMATION
SYSTEM$EXPLAIN_PLAN_JSON
TABLE_STORAGE_METRICS view

Domain 4.0: Data Governance

4.1 Monitor data.

- Apply object tagging and classifications
- Use data classification to monitor data
- Manage data lineage and object dependencies
- Monitor data quality

4.2 Establish and maintain data protection.

- Use Horizon Catalog to share and federate data outside of Snowflake
- Implement column-level security
  - Use in conjunction with Dynamic Data Masking
  - Use in conjunction with external tokenization
  - Use projection policies
- Use data masking with Role-Based Access Control (RBAC) to secure sensitive data
- Explain the options available to support row-level security using Snowflake row access policies
  - Use aggregation policies
- Use DDL to manage Dynamic Data Masking and row access policies
- Use best practices to create and apply data masking policies
- Use Snowflake Data Clean Rooms to share data
  - Clean room UI
  - Snowflake developer APIs

Domain 4.0 Study Resources

Additional Assets

Snowflake RBAC Security Prefers Role Inheritance to Role Composition (blog)

Snowflake Documentation Links

ALTER DYNAMIC TABLE
Continuous Data Protection
CREATE MATERIALIZED VIEW
Data access
Managing Governance in Snowflake
Managing Security in Snowflake
Object tagging
Projection policies

Domain 5.0: Data Transformation

5.1 Define User-Defined Functions (UDFs) and outline how to use them.

- Snowpark UDFs (for example, Java, Python, Scala)
- Secure UDFs
- SQL UDFs
- JavaScript UDFs
- User-Defined Table Functions (UDTFs)
- User-Defined Aggregate Functions (UDAFs)

5.2 Define and create external functions.

- Secure external functions
- Work with external functions

5.3 Design, build, and leverage stored procedures.

- Snowpark stored procedures
- SQL Scripting stored procedures
- JavaScript stored procedures
- Transaction management

5.4 Handle and transform semi-structured data.

- Traverse and transform semi-structured data to structured data
- Transform structured data to semi-structured data

5.5 Handle and process unstructured data.

- Use unstructured data
  - URL types
- Use directory tables
- Use the Rest API
- Use semantic views
- Use Snowflake Cortex features to:
  - Automate data categorization
  - Extract multimedia data
  - Perform semantic data analysis
  - Run text analytics in data pipelines
  - Run multimodal AI workflows within SQL queries
- Use Cortex LLM for cost management

5.6 Implement and manage development workflows and code management.

- Snowsight Workspaces and development environments
  - Snowflake Notebooks
- Git integration and version control
- Snowflake dbt Projects management
- Code deployment pipelines
- Testing and validation frameworks
- Environment management strategies

5.7 Use Snowpark for data transformations.

- Understand Snowpark architecture
- Query and filter data using the Snowpark library
- Perform data transformations using Snowpark (for example, aggregations)
- Manipulate Snowpark DataFrames

Domain 5.0 Study Resources

Additional Assets

Best Practices for Managing Unstructured Data (white paper)

Snowflake Documentation Links

ARRAY_AGG
CREATE API INTEGRATION
CREATE EXTERNAL FUNCTION
CREATE PROCEDURE
Databases, Tables & Views
External Functions
Handler writing in Snowpark
OBJECT_CONSTRUCT
Queries
Semi-Structured Data
Snowflake Scripting
Snowpark API
Stored Procedures
Transactions
TRY_PARSE_JSON
UDFs (User-Defined Functions)
Unstructured data
Working with DataFrames

SNOWPRO ADVANCED: DATA ENGINEER SAMPLE QUESTIONS

1. A Data Engineer needs to reload data into a table from files that have been in cloud storage; some files have been stored more than 64 days.

   How should the data be loaded to avoid duplicating data that is already stored in the table?

   A. Execute a `COPY INTO <table>` command with the `FORCE` option set to `= TRUE`.
   B. Delete all rows from the table and then execute a `COPY INTO <table>` command.
   C. Execute a `COPY INTO <table>` command with the `LOAD_UNCERTAIN_FILES` option set to `= TRUE`.
   D. Truncate the table and then execute a `COPY INTO <table>` statement.

2. A Data Engineer discovered that 9 days ago an outage caused a failure of a data pipeline run that was supposed to deliver data summarizing daily sales. The data did not update correctly and data is missing. The company is using the Snowflake Enterprise edition, with all data protection settings at the default values.

   The Engineer needs to verify that the pipeline architecture is compliant with these data protection rules for data in the `FENIX_DWH` database:

   1. 10 days of data history must be accessible in the SALES schema
   2. 1 day of data history must be accessible in all other schemas

   How can these requirements be met MOST cost effectively?

   A. Use the default Snowflake settings for data protection.
   B. Use the `ALTER ACCOUNT SET DATA_RETENTION_TIME_IN_DAYS = 10;` parameter.
   C. Use the `ALTER SCHEMA SALES SET DATA_RETENTION_TIME_IN_DAYS = 10;` parameter.
   D. Use the `ALTER DATABASE FENIX_DWH SET DATA_RETENTION_TIME_IN_DAYS =10;` parameter.

3. A Data Engineer wants to parse JSON in Snowflake and runs the following query:

```
create or replace temporary table vartab (id integer, v
varchar);

insert into vartab (id, v) values
     (1, '[-1, 12, 289, 2188, false,]'),
     (2, '{ "x" : "abc", "y" : false, "z": 10} '),
     (3, '[ "x" : "def", "y" : true, "z": 11 '),
     (4, '[-1, 12, 289, 2188], NULL');

 select id, try_parse_json(v)
 from vartab
 order by id;
```

What will be the result of the query?

A. The following error message will be returned: Error parsing JSON: incomplete object value.

B.

1	[-1, 12, 289, 2188, false, undefined]
2	{ "x": "abc", "y": false, "z": 10 }
3	NULL
4	NULL

C.

1	[-1, 12, 289, 2188, false, undefined]
2	{ "x": "abc", "y": false, "z": 10 }
3	[3, '["x" : "def", "y" : true, "z": 11]
4	NULL

D.

1	[-1, 12, 289, 2188, false, undefined]
2	{ "x": "abc", "y": false, "z": 10 }
3	NULL
4	[-1, 12, 289, 2188], NULL

4. A task was created to refresh data that is used to generate reports about data in a table. The task was running well until a large amount of historical data was added to the table.

   Recently, queries to generate reports are failing. The ETL team checked the task history and reports:

   1. The task has been executed successfully.
   2. Once initiated, the task resumes correctly.
   3. The task has all required grants.
   4. This is a stand-alone task with no dependencies.
   5. The task is not using a stream.

   Why is this task failing and what can be done to address the problem?

   A. The task is limited to a default duration of 60 minutes. The task needs to be recreated with the value `USER_TASK_TIMEOUT_MS` set to a higher value.
   B. The task is limited to a default duration of 120 minutes.The task needs to be recreated with the `USER_TASK_MINIMUM_TRIGGER_INTERVAL_IN_SECONDS` value set higher.
   C. The task is limited to a default duration of 60 minutes. The task needs to be recreated with the `USER_TASK_MINIMUM_TRIGGER_INTERVAL_IN_SECONDS` value set higher..
   D. The task is limited to a default duration of 120 minutes.The task needs to be recreated with the `USER_TASK_TIMEOUT_MS` value set higher.

5. A Data Engineer recreated the table, `SALES`, using the `CREATE [ OR REPLACE ] TABLE` command without backing up the existing table. The Engineer found this information in the `QUERY HISTORY` view in the `INFORMATION SCHEMA`:

   ```
   statement identifier for the "create or replace table" command:
   1e5d0ca9-003e-22e6-b858-a7f5b37c5729
   Time of execution: Wed, 26 Jan 2026 09:20:00 -0700
   ```

   How can the original table data be retrieved?

   A. ```
      DELETE FROM SALES;
      INSERT INTO SALES SELECT * FROM SALES AT (TIMESTAMP =>
      'Wed, 26 Jan 2026 09:20:00 -0700'::timestamp_tz);
      ```
   B. ```
      UNDROP TABLE SALES;
      DELETE FROM SALES;
      INSERT INTO SALES SELECT * FROM SALES AT (OFFSET =>
      -60*5);
      ```
   C. ```
      ALTER TABLE SALES RENAME TO SALES_new;
      UNDROP TABLE SALES;
      ```
   D. ```
      ALTER TABLE SALES RENAME TO SALES_new;
      UNDROP TABLE SALES;
      INSERT INTO SALES SELECT * FROM SALES BEFORE(STATEMENT =>
      '1e5d0ca9-003e-22e6-b858-a7f5b37c5729');
      ```

Correct responses for sample questions:

1: d, 2: c, 3: b, 4: a , 5: c

NEXT STEPS

REGISTERING FOR YOUR EXAM

When you are ready to register for the exam navigate here to get started. Select the exam you want to take and click "Register Now." This will take you to our Certification Management system where you will register to take the exam.

MAINTAINING YOUR CERTIFICATION

All Snowflake Certifications expire two (2) years after your certification issue date.

SnowPro Certifications can now be recertified through the Snowflake Continuing Education (CE) program which includes these options:

- Completion of eligible Snowflake Instructor Led (ILT) Training Courses
- Earning of an equivalent or higher-level SnowPro Certification

Note: You must have a valid SnowPro Certification to participate in the Continuing Education (CE) program.

The information provided in this study guide is provided for your purposes only and may not be provided to third parties.

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